

## Use dplyr and tidyverse to Complete the Same Steps (from Base R Exercise 2)

Overview: We will modify the “big data” prison inmate count dataframe and then aggregate it to the state-level. Then merge it with census populations data to create an incarceration rate.

# 1. Load both the tidyverse library and haven (to import an SPSS file). If either tidyverse or haven are not installed. Please click on “packages” and “install” before loading the library.

```
library(tidyverse)
library(haven)
```

#2. Create a dataframe named “prison\_counts” by importing the SPSS (.sav) file “ncrp\_3years\_year\_end.sav” from OneDrive.

#3. Filter out unwanted years so that only data from the year 2010 remains.

#4. Convert all variable names to lower case.

#5. Create a variable called “year” by assigning it the value of the rptyear variable.

#6. Create a dummy variable (inmate\_dummy) that is equal to “1” on each row, to make inmate counting easier. Keep only the needed variables (year, dummy, state) using SELECT function.

#7. Aggregate all need variables using the AGGREGATE function.

#8. Rename or change the name of “inmate\_dummy” to “inmate\_count” using vector or bracket syntax [ ] and the NAMES function.

#9. Import the census population data from OneDrive “census\_panel.csv”

#10. Merge the 2010 state-level inmate count data with census population data (using the Base R inner join function. Verify that the correct year has been merged.

```
#Example: database_inner <- merge(data_A, data_B, by = "ID")
```

#11. Create an incarceration rate (per 100,000 residents) for each state.

#12. Sort the (from high to low incarceration\_rate) using the order function, then print out the incarceration rate data with (including state name and year).

Example: print(dataframe).

#13. Submit your code and final output via Blackboard.